Kirtan Patel

+1 630-720-2202 | kirtannpatel2003@gmail.com | Portfolio | LinkedIn | GitHub

Summary

MS Artificial Intelligence student at San Jose State University and CS graduate (University of Illinois Chicago, May 2026) with end-to-end experience across full-stack development, applied machine learning, data engineering, and systems programming. Built production-style pipelines on 500K–1M+ record datasets, an AI-powered code review platform with LLM safety checks, and a sub-10ms C++ networking gateway. Shipped REST APIs and Docker microservices during a software engineering internship at a cybersecurity startup.

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, Go, C/C++, SQL, R
AI/LLM & Agents: OpenAI API, Anthropic API, Hugging Face, LangChain, Prompt Engineering, RAG, AI Agents, Claude Code, Cursor, Ollama
ML & Data Engineering: PyTorch, XGBoost, Neural Networks, PySpark, Apache Kafka, Apache Airflow, dbt, ETL/ELT, Pandas, NumPy
Frontend & Backend: React, Next.js, FastAPI, Node.js, REST APIs, GraphQL, WebSockets, Spring Boot
Databases & Infra: PostgreSQL, Redis, MongoDB, SQLite, Vector Databases, Docker, Kubernetes, AWS, GCP, CI/CD
Systems Programming: Multithreading, Socket Programming, Memory Management, GDB, Valgrind, Linux

Experience

Software Engineer Intern Nov 2025 – Mar 2026
Resilience Inc (Series B, cybersecurity startup)
- Diagnosed and resolved memory leaks and async race conditions in production backend systems, reducing error reports by 40%
- Improved REST API performance reducing peak latency by 30% – leveraged Claude Code, Cursor, and AI coding assistants to accelerate development from prototyping through testing and documentation
- Containerized and maintained microservices using Docker on Linux ensuring consistent deployments across environments

Teaching Assistant – Computer Science (C/C++, MATLAB) Aug 2024 – May 2026
University of Illinois Chicago
- Led C/C++ and MATLAB lab sessions for 50+ students, helping debug code and understand core programming concepts
- Reviewed student code and discussed design decisions, improving correctness, readability, and performance

Projects

Healthcare Claims Fraud Detection | *Python, XGBoost, SHAP, FastAPI, Streamlit, Docker* GitHub
- Designed scalable data model on 500K+ Medicare claims – built extraction pipeline with logging, lineage tracking, versioning, and access controls; implemented SHAP explainability for regulatory compliance
- Deployed FastAPI + Streamlit dashboard surfacing fraud risk insights to non-technical stakeholders

AI Code Review Platform | *React, FastAPI, PostgreSQL, Docker, Ollama, Anthropic API* GitHub
- Built full-stack developer tool integrating an LLM inference pipeline (Ollama/Anthropic API) with prompt engineering for line-level code analysis and quality scoring
- Implemented AI safety checks validating model outputs before surfacing results to users; used Claude Code and GitHub Copilot throughout, critically reviewing all AI-generated output before shipping

Realtime AI Collab Editor | *Next.js, TypeScript, Yjs CRDT, Supabase, WebSockets, Ollama* GitHub
- Built real-time collaborative interface with Next.js/TypeScript frontend – multi-user sync, live cursor sharing, and presence tracking; serving concurrent users with sub-100ms latency
- Implemented autosave, snapshot recovery, and incremental sync for resilience under dropped connections; integrated Ollama for on-device LLM inference

Adaptive Load-Shedding API Gateway | *C++, Linux, Sockets, HTTP, Concurrency* GitHub
- Built C++ gateway from scratch using OS-level primitives – tracked P95 latency maintaining sub-10ms response at 3x normal traffic with zero-error correctness guarantees under sustained burst
- Diagnosed race conditions using GDB/Valgrind; validated correctness with automated integration testing and CI/CD pipeline on Linux

Education

San Jose State University Expected May 2028
MS in Artificial Intelligence
University of Illinois at Chicago May 2026
BS in Computer Science *Dean's List*
- **Coursework:** Distributed Systems, OS, Database Systems, Algorithms, Data Structures, Statistics, Computer Networks